ACCESCO LIVING PRIVATE LIMITED

CIN: U47912WB2025PTC285309

ISO 9001:2015 CERTIFIED
DPIIT Startup India Certified

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# The Household Consumption Engine

Wardrobe as a calculated AI feature, not a product. Plus the No-Expiry Engine on the admin side. Working draft for discussion, 30 October 2026. Not approved for external use.

**The household never logs anything. Every item accepted at the door, every kit, every meal and every cook visit teaches the engine how fast that home uses each item. From that, it estimates what is running low and offers one tap to reorder.**

## 1. What changes

TRACO and Grokly Wardrobe are removed as products. There is no planning tool and no pantry to maintain. In their place sits one calculated feature inside the app, still called Wardrobe, that works entirely from data the store already has: accepted deliveries, kit orders, meal orders (Serves 1 or Serves 2) and cook visits. It shows the household what is likely running low and lets them reorder in one tap. Nothing is ever ordered automatically.

The No-Expiry Engine stays, but on the admin side only: it governs which batch is sold, which goes to the kitchen and which is returned, and it never asks the customer to do anything.

## 2. The consumption rate of a household for an item

For household h and item i, let the accepted quantities (at the door) be q_1 ... q_n at dates d_1 ... d_n. Kits and cook visits count as accepted quantities of their ingredients through the recipe; cooked meals do not consume the home's stock and are excluded. The raw rate is total quantity divided by the days it covered:

**Argha's Formula**

$$r_{raw}(h, i) = \sum_{j=1}^{n-1} q_j \,/\, (d_n - d_1)$$

The last pack is excluded from the numerator because it is still being used. With few purchases this estimate is noisy, so it is shrunk toward the catchment average rate for that item, rho(i), with strength k (in days of evidence):

**Argha's Formula**

$$r(h, i) = (k \times \rho(i) + \sum q_j) \,/\, (k + (d_n - d_1))$$

A new household therefore starts at the neighbourhood's rate and moves to its own as evidence accumulates. Prior strength k is set per category; a starting value is 14 days for fresh items and 30 days for staples.

Serves 1 and Serves 2 choices feed the same estimate: the share of Serves 2 orders is used as a soft signal of household size when rho(i) is chosen (one-person and two-plus-person catchment averages).

## 3. Estimated stock and days to low

The stock at home for item i is estimated from the last accepted quantity q_n, the time since it arrived, and the rate:

**Argha's Formula**

$$Stock(h, i, t) = \max(0, q_n - r(h, i) \times (t - d_n))$$

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The item is flagged as running low when the estimated stock falls to a fraction theta of the pack (starting prior theta = 0.20). The expected day is:

| Argha's Formula |
| --- |
| **T_low(h, i) = d_n + ( q_n x (1 - theta) ) / r(h, i)** |

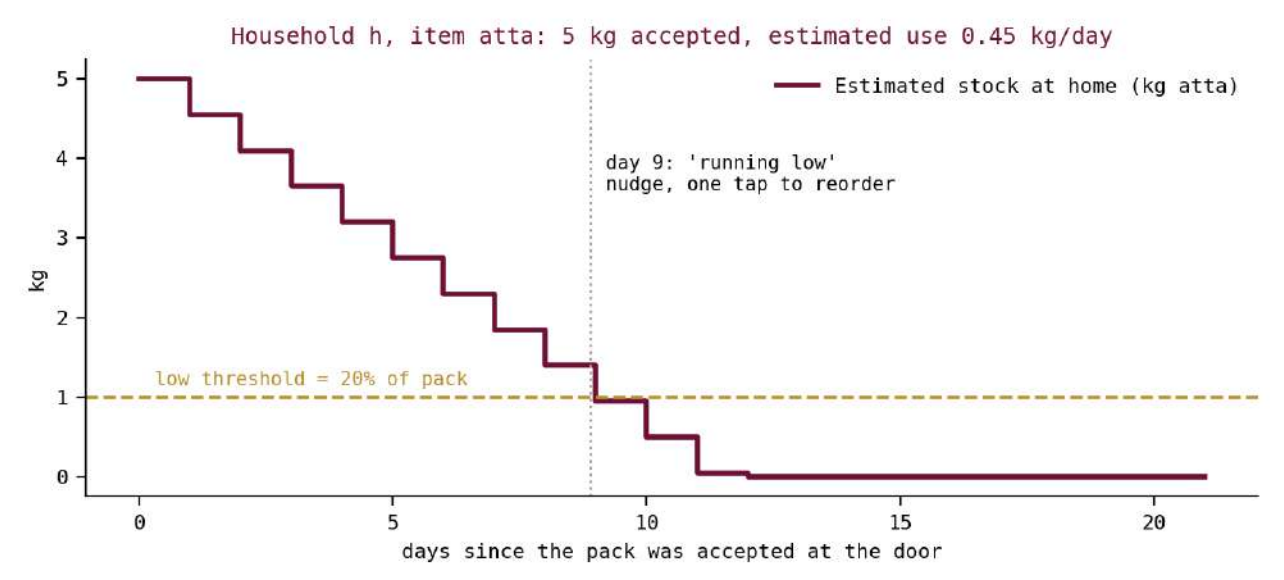



Figure 1. A 5 kg pack of atta accepted at the door, used at an estimated 0.45 kg a day. The engine flags it as running low on day 9 and offers one tap to reorder. If the household reorders earlier or later, the rate is updated.

## 4. Reorder probability

Rates alone miss habit: some homes buy weekly whatever the pack size. So the engine also learns each household's inter-purchase intervals for the item, delta_1 ... delta_m, and fits them to a gamma distribution (shape and scale from the sample mean and variance, shrunk toward the catchment's values with the same k). The probability that the household reorders by day t is the fitted cumulative distribution:

| Argha's Formula |
| --- |
| **P_reorder(h, i, t) = Gamma_cdf( t - d_n ; shape(h, i), scale(h, i) )** |

The nudge appears when either signal says so: stock is estimated low, or P_reorder passes 0.5. The nudge is a single line in the app with one tap; it is never an automatic order.

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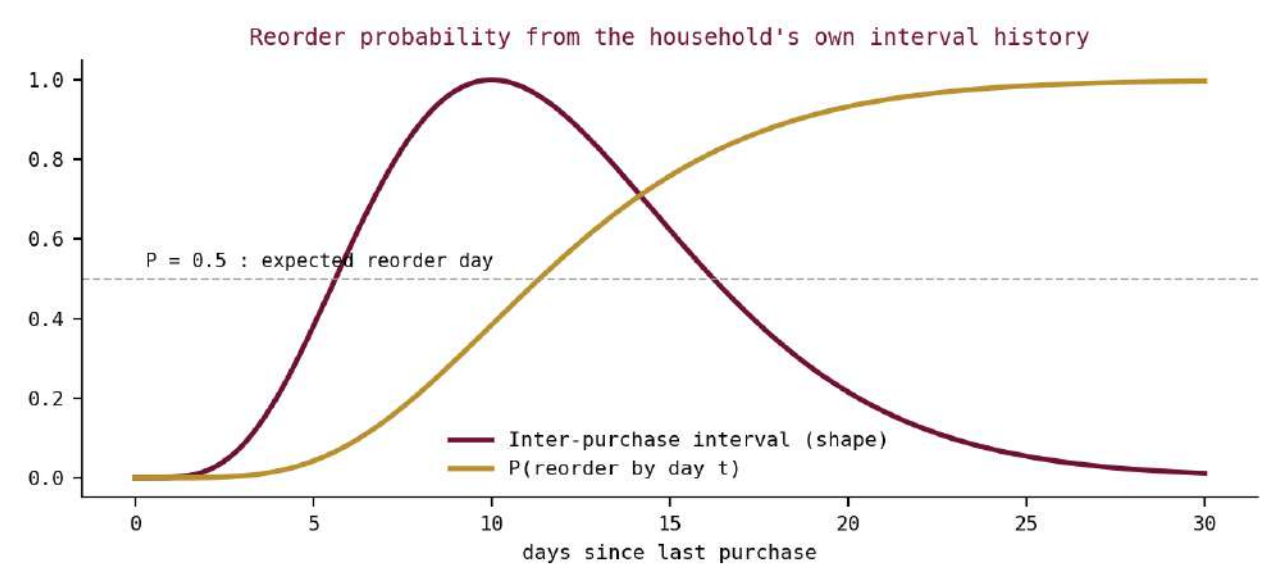



Figure 2. The household's own interval history (shape) and the resulting reorder probability. The day the curve crosses 0.5 is the expected reorder day.

## 5. How the engine learns from the door

| Event | What the engine updates |
|---|---|
| Item accepted at the door (Doorstep Check) | q_n and d_n for that item; the rate r(h, i) is recomputed. |
| Item rejected at the door | Nothing in the household's stock; the rejection goes to the store's quality data (R(i) in Pull Theory). |
| Ingredient kit or cook visit | The recipe's ingredient quantities are added as accepted stock. |
| Cooked meal (Serves 1 or 2) | No stock change; the size choice updates the household-size signal. |
| Household reorders earlier than predicted | Rate rises; interval history shortens. |
| Household ignores a nudge | Rate falls slightly; the nudge is suppressed for that item for a set number of days. |

Privacy: all of this is household purchase data under India's data protection law. Consent is taken once at sign-up, the data is used only to power this feature and the store's Pull calculation in aggregate, and the household can delete it on request.

## 6. What the engine gives back to the store

Summing each household's expected reorders for tomorrow gives the store a second, independent estimate of demand for item i. In Pull Theory it enters as a check on A(i, w): where the household sum and the weekday average disagree by more than a set margin, the larger is used for fresh items (the kitchen absorbs the difference) and the average for packaged goods.

| Argha's Formula |
|---|
| **E_h(i, t+1) = SUM over households h of P_reorder(h, i, t+1) x q_typical(h, i)** |

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## 7. The No-Expiry Engine (admin side)

Every batch that enters the store carries a manufacture or harvest date and an expiry or best-before date. The engine computes a remaining life fraction for the batch each day:

**Argha's Formula**

**L(batch, t) = ( expiry - t ) / ( expiry - manufacture )**

Three thresholds govern what the batch may do. Starting priors for packaged goods: L_sell = 0.50, L_cook = 0.30, below which the batch is returned or composted. Fresh produce uses days instead of fractions: sell within 48 hours of harvest, cook within 72, then compost. Both are tightened by category and by the FSSAI floor (no item leaves the store for a customer with less than 30% of its life or 45 days remaining).

| Remaining life L | What the batch may do | Who acts |
|---|---|---|
| L >= L_sell | Sellable on the shelf. Picking is FEFO: the oldest sellable batch is picked first. | Picker, guided by the admin panel |
| L_cook <= L < L_sell | Not for the shelf. Fresh Routing moves it to the kitchen for the same day's cooking. | Kitchen lead |
| L < L_cook | Returned to the supplier where terms allow, otherwise compost or recycling partner. | Store manager |

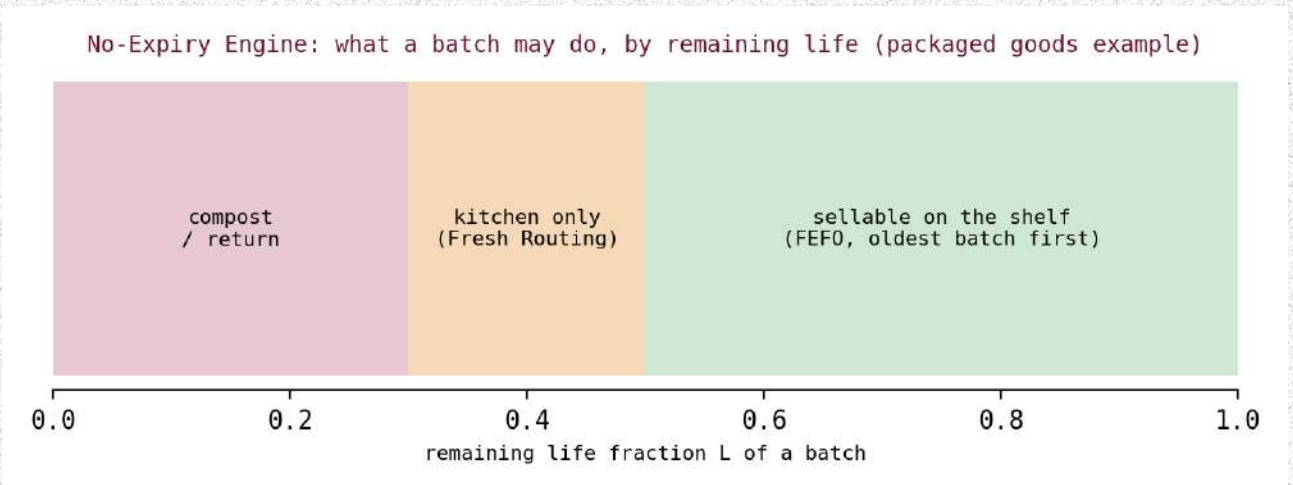



Figure 3. The three bands of the No-Expiry Engine for a packaged-goods batch. The customer sees only the result: nothing arrives with less than the promised life.

Customer-facing promise that follows from it: every item delivered carries at least the promised remaining life for its category, and Doorstep Check lets the household verify it. The engine is the reason the promise can be kept at scale without inspecting every order by hand.

## 8. What to measure weekly

| Metric | Target | Why |
|---|---|---|
| Nudge accuracy: nudged items reordered within 3 days | above 60% | The engine should be right more often than not before it earns more prominence. |
| Nudge acceptance rate (one-tap reorders) | rising week on week | Measures whether the feature is useful, not noisy. |
| Households with at least one accurate rate after 4 weeks | above 80% | Shows the shrinkage prior converging to real behaviour. |

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| Metric | Target | Why |
|---|---|---|
| Items delivered below the promised life | zero | The No-Expiry promise. |
| Batches moved to kitchen by Fresh Routing | tracked | Feeds Pull Theory's absorption numbers. |

**Summary: Wardrobe becomes a calculated feature that asks the household for nothing and offers one useful tap; the No-Expiry Engine runs the shelf from the admin side; and both feed the store's Pull calculation. No planning tools, no pantry chores, no jobs for the customer.**